

Q1: What is a degree of freedom of the system of rigid bodies - Solution:-

Answer The number of coordinates required to specify the position of a system of one or more particles is called the number of degree of freedom of the system.

EXAMPLE:- A particle moving freely in space $\mathbb{R}^3$ requires 3 coordinates e.g. (x, y, z) , to specify its position. Thus the number of degree of freedom is 3.

Q2:- Find the moment of Inertia of a ring of radius a about an axis through centre.

Solution:-

Let M be the mass and a the radius of the ring.

Then the mass per unit length will be

$M/2\pi a$. we regard this ring to be composed of small elements of mass (δm) each of length δs , we can write it as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi a} \Rightarrow \delta m = \frac{M}{2\pi a} \delta s$$

moment of Inertia of the element about an axis through centre O and the perpendicular to the plane of the ring

$$I_C = \frac{1}{2} Ma^2 \quad \text{--- (2)}$$

By substituting eq (2) in eq (1), we have
Taking Ma^2 common we get

$$I_A = Ma^2 \left(\frac{1}{2} + 1 \right)$$

$$I_A = \frac{3}{2} Ma^2 \quad \text{Answer -}$$

Q22: Write different examples of conservative Forces.

- Sol: ① Gravitational force is an example of a Conservative force -
- ② Elastic Spring force is an example of Conservative force -
- ③ The work done of a Particle moving along a closed path is zero and the force which causes such motion is Conservative -

Q23: State the "Chasles' Theorem".

Sol: Chasle's theorem states that the most general rigid body displacement can be produced by a translation along a line (called its Screw axis) followed (or preceded) by a rotation about the line -

Infinitesimal Rotations can be represented by vectors.

Q20 Particle of masses $2m$, $3m$ & $4m$ are held in rigid light framework at points $(1,0,1)$, $(1,1,0)$ and $(0,1,1)$. Calculate moment of Inertia of x-axis.

Sol Given $m_1=2$ $m_2=3$ $m_3=4$

Points are $P_1(1,0,1)$ $P_2(1,1,0)$, $P_3(0,1,1)$

Moment of Inertia about x-axis is

$$I_{xx} = \sum_i m_i (y_i^2 + z_i^2)$$

$$I_{xx} = m_1 (y_1^2 + z_1^2) + m_2 (y_2^2 + z_2^2) + m_3 (y_3^2 + z_3^2)$$

$$I_{xx} = 3(0+1) + 5(1+0) + 7(1+0)$$

$$I_{xx} = 3 + 5 + 7$$

$$I_{xx} = 15 \text{ Answer}$$

Q21:- Use the parallel axis theorem to find the moment of Inertia of a Solid Circular Cylinder about a line on the Surface of the cylinder and parallel to axis of cylinder-

sol:- By Parallel axis theorem

$$I_A = I_c + Ma^2 \text{ --- (1)}$$

Since I_c which is the moment of Inertia of a Solid Circular cylinder about an axis passing from the Centre of mass is defined

Q18 If the surface area of the ring is $2\pi r dr$ and surface area of annulus is $\pi(R_2^2 - R_1^2)$ then find the mass dm of ring. Whereas mass of disc is M .

Sol: - The surface area of the ring is

$$\text{Area} = (2\pi r) dr = 2\pi r dr$$

Since the surface area of the annulus is

$$\pi(R_2^2 - R_1^2)$$

Therefore, we can

$$\frac{dm}{M} = \frac{2\pi r dr}{\pi(R_2^2 - R_1^2)}$$

$$dm = \frac{2\pi r dr}{(R_2^2 - R_1^2)} M \quad \text{Required Sol}$$

Q19 - State Euler's theorem -

Sol: - "A rotation of a rigid body about a fixed point of the body is equivalent to rotation about a line which passes through the point."

The line referred to is called the instantaneous axis of rotation. Rotations can be considered as finite or infinitesimal. Finite rotation can't be represented by vectors since the commutative law fails. However

Q16 What is the impulse developed by mass of 100g which changes its velocity from $\hat{i} + \hat{j} + \hat{k}$ to $2\hat{i} + 2\hat{j} + 3\hat{k}$ m/sec?

$$m = 100 \quad v_1 = \hat{i} + \hat{j} + \hat{k}, \quad v_2 = 2\hat{i} + 2\hat{j} + 3\hat{k}$$

we know that

$$\text{Impulse} = P_2 - P_1$$

$$= mv_2 - mv_1$$

$$= 100 \{ (2\hat{i} + 2\hat{j} + 3\hat{k}) - (\hat{i} + \hat{j} + \hat{k}) \}$$

$$= 100 (2\hat{i} - \hat{i}) + (2\hat{j} - \hat{j}) + (3\hat{k} - \hat{k})$$

$$= 100 (\hat{i} + \hat{j} + 2\hat{k}) \quad \text{Ans}$$

Q17 if a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{j} - 3\hat{j} + 5\hat{k}$ relative to fixed

Coordinate system, and $\vec{A} = \sin t \hat{i} - \cos t \hat{j} + e^t \hat{k}$

then find $d\vec{A}/dt$ w.r.t the fixed point system

where $\left(\frac{d\vec{A}}{dt}\right) = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-2t} \hat{k}$ and

$$\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t) \hat{i} - (-2e^{-t} + 3\sin t) \hat{j} + (\cos t + 2\sin t) \hat{k}$$

$$\text{Sol:- } \left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \vec{\omega} \times \vec{A}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-2t} \hat{k} + (4e^{-2t} - 3\cos t) \hat{i} - (-2e^{-t} + 3\sin t) \hat{j} + (\cos t + 2\sin t) \hat{k}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = (-\sin t + 4e^{-2t} - 3\cos t) \hat{i} + (\cos t + 2e^{-t} - 3\sin t) \hat{j} + (-2e^{-2t} + \cos t + 2\sin t) \hat{k}$$

Required Answer

Q15 Consider a small element of a uniform elliptic plate has moment of Inertia $I = \delta m y^2$ write down the expression for moment of Inertia of whole plate (Note do not solve the expression).

$$I_x = \frac{1}{4} a b^3 \pi \times \frac{M}{\pi a b} = \frac{1}{4} M b^2$$

where for elliptical plate we have

$$\rho = \frac{M}{\pi a b}$$

is required expression for $M \cdot I$

OR

$$I_{xx} = \int (\delta m) y^2 = \int_{\text{plate}} \rho d s y^2 \quad \text{Ans}$$

Q14 Find the moment of Inertia of a Solid Circular Cylinder of radius a , mass M and the height of the cylinder h about the axis of the cylinder. whereas moment of Inertial of one Shell be

$$I_{\text{shell}} = r^2 \frac{2M}{a^2} r dr = \frac{2M}{a^2} r^3 dr$$

(the mass of one shell is $\frac{2M}{a^2}$, height is same as h , thickness be dr and the radius be r and)

Sol:— $I_{\text{shell}} = r^2 \frac{2M}{a^2} r dr = \frac{2M}{a^2} r^3 dr$

Thus the total moment of Inertia of the Solid Circular cylinder will be

$$I_{\text{cylinder}} = \int_{r=0}^a I_{\text{shell}}$$

$$I_{\text{cylinder}} = \int_{r=0}^a \frac{2M}{a^2} r^3 dr$$

$$= \frac{2M}{a^2} \int_0^a r^3 dr$$

$$= \frac{2M}{a^2} \left| \frac{r^4}{4} \right|_0^a = \frac{2M}{a^2} \frac{a^4}{4}$$

$$I_{\text{cylinder}} = \frac{1}{2} M a^2 \text{ is the M.I. of solid cylinder.}$$

Answer

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$$2\pi\rho a^5 \int_0^{\pi/2} \frac{3\sin\theta - \sin 3\theta}{4} d\theta \Rightarrow (2\pi\rho \frac{a^5}{3}) \frac{1}{4} [-3\cos\theta - \frac{\cos 3\theta}{3}]_0^{\pi/2}$$

$$\pi\rho \frac{a^5}{10} [(0) - (-3 + 1/3)]$$

Q13. If $e_p = \cos\phi \hat{i} + \sin\phi \hat{j}$ and $e_\phi = -\sin\phi \hat{i} + \cos\phi \hat{j}$ then Prove that $\frac{de_p}{dt} = \dot{\phi} e_\phi$ and $\frac{de_\phi}{dt} = -\dot{\phi} e_p$

where dots denote differentiation w.r.t to time t.

Sol:- We have $e_p = \cos\phi \hat{i} + \sin\phi \hat{j}$
 $e_\phi = -\sin\phi \hat{i} + \cos\phi \hat{j}$
 Then $\frac{de_p}{dt} = \frac{d}{dt} (\cos\phi \hat{i} + \sin\phi \hat{j})$
 $= \frac{d\cos\phi}{dt} \hat{i} + \frac{d\sin\phi}{dt} \hat{j} = \dot{\phi} (-\sin\phi \hat{i} + \cos\phi \hat{j})$
 $= \dot{\phi} e_\phi$

$$\frac{de_\phi}{dt} = \frac{d}{dt} (-\sin\phi \hat{i} + \cos\phi \hat{j})$$

$$= -\cos\phi \dot{\phi} \hat{i} - \sin\phi \dot{\phi} \hat{j}$$

$$= -\dot{\phi} (\cos\phi \hat{i} + \sin\phi \hat{j}) = -\dot{\phi} e_p$$

Next $\frac{de_\phi}{dt} = -\dot{\phi} e_p$

By using relation for density

$$\rho = \frac{M}{\frac{2}{3}\pi a^3}$$

we obtain

$$I_{zz} = \frac{2}{5} Ma^2$$

which is required expression for mom of Inertia of hemisphere.

Q11 Consider the Conservative force field $F = -kr^3 \vec{r}$. Find the Potential energy of the given force field.

Ans As $F = -kr^3 \vec{r}$

$$F = -\nabla V$$

$$\begin{aligned} F = -kr^3 \vec{r} &= -k(x^2 + y^2)^{\frac{3}{2}} (x\vec{i} + y\vec{j}) \\ &= -k(x^2 + y^2)^{\frac{3}{2}} x\vec{i} - k(x^2 + y^2)^{\frac{3}{2}} y\vec{j} = -\nabla V \\ &= -\frac{\delta V}{\delta x} \vec{i} - \frac{\delta V}{\delta y} \vec{j} - \frac{\delta V}{\delta z} \vec{k} \end{aligned}$$

By comparing we get

$$\frac{\delta V}{\delta x} = k(x^2 + y^2)^{\frac{3}{2}} x, \quad \frac{\delta V}{\delta y} = k(x^2 + y^2)^{\frac{3}{2}} y$$

$$\text{and } \frac{\delta V}{\delta z} = 0$$

from which $\frac{\delta V}{\delta z}$, by omitting the constant, we get

$$V = \frac{1}{5} k(x^2 + y^2)^{\frac{5}{2}} = \frac{1}{5} kr^5$$

is Required potential.

Q12 Write down the expression for moment of Inertia of a uniform hemisphere of radius a about its axis of symmetry, where $dv = dr(r d\theta)(r \sin \theta d\phi) = r^2 \sin \theta dr d\theta d\phi$

$$dv = dr(r d\theta)(r \sin \theta d\phi) = r^2 \sin \theta dr d\theta d\phi$$

$$\begin{aligned} \text{Therefore } I_{zz} &= \rho \int_0^a \int_0^{\pi/2} \int_0^{2\pi} r^4 \sin^3 \theta dr d\theta d\phi \\ &= \int_0^a r^4 dr \int_0^{\pi/2} \sin^3 \theta d\theta \int_0^{2\pi} d\phi \end{aligned}$$

$$= 2\pi \rho \frac{a^5}{5} \int_0^{\pi/2} \sin^3 \theta d\theta$$

Q10 Calculate the moment of Inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point.
(let the x -axis be chosen along the length of the rod, with origin at one end point let M and a be the mass and length of rod respectively and suppose the rod to be composed of small elements.)

Ans let dm and dx be the mass and the length of the specific element of the rod at a distance x from the end point O .

Then
$$\frac{dm}{dx} = \frac{M}{a}$$

$$dm = \frac{M}{a} dx$$

Then the moment of Inertia of this element about the given axis is

$$I_{\text{element}} = dm x^2 = \frac{M}{a} x^2 dx$$

Hence the moment of Inertia of the whole rod will be

$$\begin{aligned}
 I &= \sum_{\text{all elements}} \frac{M}{a} x^2 dx \\
 &= \frac{M}{a} \int_0^a x^2 dx \\
 &= \frac{M}{a} \left| \frac{x^3}{3} \right|_0^a = \frac{M}{a} \frac{a^3}{3} \\
 &= \frac{1}{3} Ma^2 \quad \text{Ans}
 \end{aligned}$$

will be

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Required Solution -

Q8 To find the Inertia matrix of uniform Solid cuboid (Parallelepiped) at one of its Corner, if $\bar{I}_{xx} = \frac{M}{3}(b^2 + c^2)$ Then find I_{yy} and $\bar{I}_{zz}$.

Sol $I_{yy} = \frac{M}{3}(b^2 + c^2)$

$$\bar{I}_{zz} = \frac{M}{3}(a^2 + b^2)$$

Q9 if Moment of Inertia of a uniform Circular Cylinder of length b and radius a about x -axis through the Centre and Perpendicular to the Central axis is $I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

$$\bar{I}_x = \bar{I}_y = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

Q6:- if $r = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$ and $P =$
 $P = -a m \omega \sin \omega t \hat{i} + b m \omega \cos \omega t \hat{j}$ then
 find the angular momentum.

Solution:-

$$\text{Angular momentum} = r \times p$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a \cos \omega t & b \sin \omega t & 0 \\ -a m \omega \sin \omega t & b m \omega \cos \omega t & 0 \end{vmatrix}$$

$$= (0)\hat{i} - (0)\hat{j} + (ab m \omega \cos^2 \omega t + ab m \omega \sin^2 \omega t)\hat{k}$$

$$\because \cos^2 \omega t + \sin^2 \omega t = 1$$

$$= ab m \omega \hat{k} \quad \text{Required Solution.}$$

Q7:- Write the expression for total energy of a particle executing Simple Harmonic Motion.

Solution:-

if T is the kinetic energy, V the potential energy and $E = T + V$ the total energy of a simple harmonic oscillator, then we have

$$K.E = T = \frac{1}{2} m v^2$$

$$V = \frac{1}{2} k x^2$$

Then the total energy of S.H.M

Q4: write down the mathematical expression for angular momentum of a single particle.

Solution: — The angular momentum of a single particle is defined as

The cross product of linear momentum and position vector of concerned particle.

mathematically

$$L = r \times mv.$$

Q5: Write the necessary and sufficient condition for a conservative force field.

Solution: — Solution: —

i A force field F is conservative if and only if there exist a continuously differentiable scalar field V such that

$$F = -\nabla V, \text{ equivalently, if and only if}$$

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = 0$$

Identically

ii A continuously differentiable force field F is conservative if and only if for any closed non-intersecting curve C (simple closed curve)

$$W = \oint_C \vec{F} \cdot d\vec{r} = 0 \quad \text{i.e. the total}$$

work done in moving a particle around any closed path is zero.

equals $\frac{M}{2\pi a} \int ds a^2$.

Therefore the M.I of the whole ring will be

$$I_{\text{ring}} = \frac{M}{2\pi a} a^2 \sum_{\text{Elements}} ds = \int ds = \frac{Ma}{2\pi} \times 2\pi a = Ma^2$$

Hence Ma^2 is the required moment of Inertia of the ring.

Q3:- what is the ~~magnitude~~ Impulse developed by a mass of 100gm which change its velocity from $i+j+k$ to $2i+2j+3k$ m/sec?

Solution:- $V_1 = i+j+k$, $V_2 = 2i+2j+3k$

$$\text{Impulse} = P_2 - P_1 \quad m = 100$$

$$= mV_2 - mV_1$$

$$m(V_2 - V_1)$$

Substituting the values, we get

$$\text{Impulse} = 100(2i+2j+3k - (i+j+k))$$

$$\text{Impulse} = 100(i+j+2k) \quad \underline{\text{Answer}}$$

Impulse magnitude self

$$= 100 \sqrt{(1)^2 + (1)^2 + (2)^2}$$

$$= 100 \sqrt{1+1+4}$$

$$= 100\sqrt{6}$$